

Human Position and Angle Detection System

AI/ML Internship — Day 8

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Module: Human Body Detection System

Introduction

Artificial Intelligence has revolutionized the fashion industry by enabling customers to visualize clothing digitally before making a purchase. Virtual Try-On (VTON) systems combine computer vision, deep learning, and image synthesis techniques to generate realistic garment fitting experiences.

Unlike traditional online shopping, Virtual Try-On technology allows users to upload their own images and preview how garments would appear on their bodies. This improves customer satisfaction, reduces product returns, and enhances personalized shopping experiences.

This research focuses on understanding the complete Virtual Try-On pipeline and the Artificial Intelligence techniques used to generate realistic clothing overlays.

Objectives

The objectives of this research are:

- Study the architecture of Virtual Try-On systems.
 - Understand the complete AI processing pipeline.
 - Analyze body detection and pose estimation methods.
 - Study garment alignment and warping techniques.
 - Explore image synthesis methods.
 - Demonstrate a simple practical implementation.
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1. Virtual Try-On Systems

1.1 Overview

A Virtual Try-On system is an AI-powered application that digitally fits clothing onto a user's image.

The primary objective is to generate realistic visualizations without requiring the customer to physically wear the garment.

Modern Virtual Try-On systems combine:

- Computer Vision

- Deep Learning
 - Pose Estimation
 - Image Segmentation
 - Garment Warping
 - Image Generation
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1.2 Importance

Virtual Try-On technology offers several advantages:

- Better online shopping experience
 - Reduced product returns
 - Personalized recommendations
 - Improved customer engagement
 - Increased confidence before purchase
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2. AI Pipeline

The Virtual Try-On process consists of several stages.

Step 1: User Image Upload

The user uploads a front-facing image.

Requirements:

- Full body visibility
 - Good lighting
 - Minimal background clutter
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Step 2: Human Detection

The AI model detects the human subject within the image.

Common techniques:

- YOLO
 - Faster R-CNN
 - MediaPipe Pose
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Step 3: Body Segmentation

Body segmentation separates:

- Hair
- Face
- Arms
- Legs
- Clothing
- Background

This helps isolate the human body for garment placement.

Step 4: Pose Estimation

Body landmarks are extracted.

Important landmarks include:

- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

Pose estimation helps determine body orientation.

Step 5: Garment Selection

The desired clothing image is selected from the product catalog.

The garment image may include:

- Shirt
 - T-Shirt
 - Jacket
 - Dress
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Step 6: Garment Alignment

The selected garment is resized and positioned according to the user's body shape.

Alignment depends on:

- Shoulder width
 - Chest position
 - Torso length
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Step 7: Garment Warping

Garment warping adjusts the clothing shape to fit different body proportions.

Popular methods include:

- Thin Plate Spline (TPS)
 - Geometric Matching Module (GMM)
 - Flow-based Warping
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Step 8: Image Synthesis

The warped garment is blended with the user image to generate the final output.

Deep learning models preserve:

- Fabric texture
 - Wrinkles
 - Shadows
 - Lighting consistency
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3. AI Fitting Systems

AI fitting systems estimate body dimensions automatically.

Measurements commonly estimated include:

- Shoulder Width
- Chest Width
- Waist Position
- Hip Width
- Arm Length

These measurements improve fitting quality.

5. Applications

Virtual Try-On systems are widely used in:

Fashion E-commerce

Digital garment fitting.

Smart Retail

Interactive shopping experiences.

Fashion Recommendation

Personalized outfit suggestions.

AR Shopping

Real-time clothing visualization.

6. Practical Implementation

Simple Clothing Overlay Demonstration

A basic OpenCV implementation was created to simulate garment placement.

Python Code

```
import cv2

person = cv2.imread("person.jpg")
shirt = cv2.imread("shirt.png")

if person is None:
    print("person.jpg not found")
    exit()

if shirt is None:
    print("shirt.png not found")
    exit()

# Resize images
person = cv2.resize(person, (600, 800))
```

```
shirt = cv2.resize(shirt, (350, 300))

# Shirt position
x = 140
y = 125

h, w, _ = shirt.shape

# Overlay shirt
person[y:y+h, x:x+w] = shirt

# Save output
cv2.imwrite(
    "virtual_tryon_output.jpg",
    person
)

print("Virtual Try-On Generated Successfully")

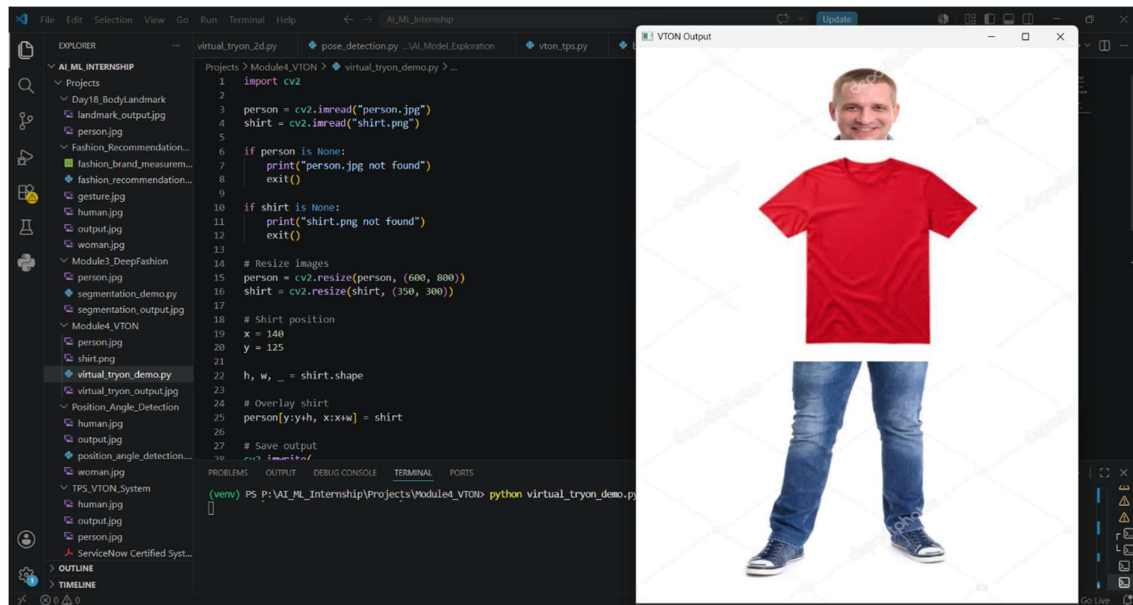
cv2.imshow(
    "VTON Output",
    person
)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Code Explanation

The implementation loads a user image and a garment image, resizes both inputs, and overlays the garment onto the body region.

Although this prototype does not use Artificial Intelligence, it demonstrates the basic concept of image composition used in Virtual Try-On systems.



Output:



7. Observations

- Accurate body localization improves fitting quality.
- Garment alignment is essential for realism.
- Image synthesis depends heavily on segmentation accuracy.
- Deep learning models outperform simple overlay techniques.
- Proper body measurements improve personalization.

8. Research Findings

The study indicates that Virtual Try-On systems require the integration of multiple AI components including body detection, segmentation, pose estimation, garment warping, and image synthesis.

Among these stages, garment warping and body segmentation contribute significantly to the realism of the final output.

Modern AI architectures have greatly improved clothing adaptation and user experience.

9. Learning Outcomes

During this research, the following concepts were studied:

- Virtual Try-On Systems
 - Computer Vision
 - Body Detection
 - Pose Estimation
 - Garment Alignment
 - Garment Warping
 - Image Synthesis
 - AI Fashion Applications
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10. Conclusion

This research explored the complete Virtual Try-On pipeline and the Artificial Intelligence techniques used to generate realistic digital garment fitting experiences. The study demonstrated how body detection, segmentation, pose estimation, and garment warping work together to create personalized virtual fitting systems.

The findings highlight the growing importance of AI-powered fashion technologies in modern e-commerce platforms and indicate that future developments in deep learning will further enhance realism, personalization, and user satisfaction.